

Emotional or Intellectual: How Do We Vote?
Examining the Voting Behavior of Virginia Beach Locals in
Relation to the Downs-Spatial and Michigan Models

Faith Grant

99/2020

In this research project, I worked in the field of political science, investigating whether the electorate makes smart decisions when they go to vote in the polls. My senior research question was, in a general sense, do people vote intellectually or emotionally. In order to measure this, I looked at two models of voting behavior: the Downs-Spatial Model/Rational Choice Theory, a model of ideological voting, and the Michigan Model, a psychological way of voting. My hypothesis was that when surveyed, more people would demonstrate the Michigan Model of Voting Behavior. I needed skills in statistical analysis and survey creation and knowledge of the main factors that influence voting. My findings were inconclusive, for the most part, as I had means to believe there was response bias.

Keywords: Human Behavior, Zingher

Abstract

In the midst of the third presidential impeachment, three years after the unpredicted victory of the 45th United States president, Donald Trump, scholars and nonscholars alike question how Donald Trump managed to win the presidency. Many scholars turn to voter turnout, which for the 2016 U.S. presidential election was reported as 138 million people, which despite the size, is a relatively small proportion of voters, only 58% of the voting-eligible population. Although the United States voter turnout is a problem knowing how the people who *do* vote make their decision is just as critical. This study works to build on the vast research of voting behavior, specifically regarding the Michigan Model of Voting Behavior and Downs-Spatial Model/Rational Choice Theory to encourage citizens not just to vote, but to vote intelligently. A survey was created and given to 50 participants from 4 different locations in Virginia Beach, Virginia meant to test assumptions of these complex models. The survey explores the assumptions that 1.) voters are rational and will choose candidates who are closest to their ideals (Downs-Spatial Model/Rational Choice Theory) or 2.) that party identification and social factors influence voters the most (Michigan Model). This study compared voter demographics with knowledge of certain policies and reactions to theoretical candidates, observing and analyzing the influence of ideology and social characteristics on voting. The survey resulted in support for the Rational Choice Theory/Downs Spatial Model, but further analysis may rule the conclusion inconclusive.

Introduction

The ability to participate in the decisions concerning a locality, state, or nation through voting is the cornerstone of American democracy. This civic duty ensures that each citizen has a say in the inner-workings of government, a government meant to serve the citizens who elect it. A vital step in the democratic process, it's important to study all aspects of voting. This is where political science, a social science focused on analyzing government and political behavior/activity, comes into play ("Political Science: Definition of Political Science by Lexico", n.d.). Often overlapping with statistics, psychology, sociology, and other sciences, political scientists have studied voting behavior, benefits, and more.

Voting behavior touches on the factors and affective influences (race, gender, religion, etc) that results in a voter's decision which in turn can make or break a candidate running for office. While these are personal factors that influence one's opinion, the factors candidates are chosen by tend to fall under public policy, government performance, and personal characteristics of the candidates which are influenced by "general attitudinal factors" of party identification and ideological orientation ("Voting Behavior", n.d.). This electoral behavior also serves as a pathway to seats as consultants, analysts, and staff of politicians. The two particular models influencing this research are the Michigan model and the Down-Spatial model paired with the Rational Choice Theory.

The Michigan model, proposed in the 1950s, is a theory of voting in which citizens vote based on sociological factors, psychological party attachment, and party identification. (Peele, 2004, p. 325). It emphasizes sociological and parental characteristics and social status as short

term factors and party loyalty as a long term effect influencing voter behavior. It's been challenged by more modern models and theories such as rational choice, retrospective voting, and prospective voting, while also serving as a base for other theories such as the party-lining voting. In the conducted survey, the socio-psychological analysis of voting behavior is believed to be the dominant model due to its touch on the ethos, as well as stubbornness, of the average American citizen.

The Downs-Spatial Model serves as a foil to the Michigan model. Critiqued as giving American voters too much credit, it assumes voters are thoroughly educated on policies and candidates and puts emphasis on ideology as well as moderatism (Vik, 2013). Found in Anthony Downs' *An Economic Theory of Democracy*, there are multifarious propositions including those to do with the median voter and vague party stance. However, for the specific research of this particular topic, one crucial assumption was targeted from this model: voters choose candidates based on the proximity of their positions and ideologies (Stokes, 1963). In this study, it's paired with the Rational Choice Theory also assessed by Anthony Downs in *An Economic Theory of Democracy* and a part of the Downs Model, that states voters make rational choices to maximize personal utility, self-interest, as opposed to the psychological attachment of a party (Farber, 2010).

Any voter would claim their intentions rest only in their ideology, but political scientists, policymakers, and campaign managers can never be so sure. With something that holds such weight in American society, isn't it important to know that voters make smart decisions instead of emotional ones? To answer this question, a survey was conducted, to determine a citizen's sociological factors in comparison to their party attachment. It was hypothesized through the

construction and analysis of contingency tables, the validity of the Michigan model would be proven as voters demonstrated discrepancies in party identity and knowledge.

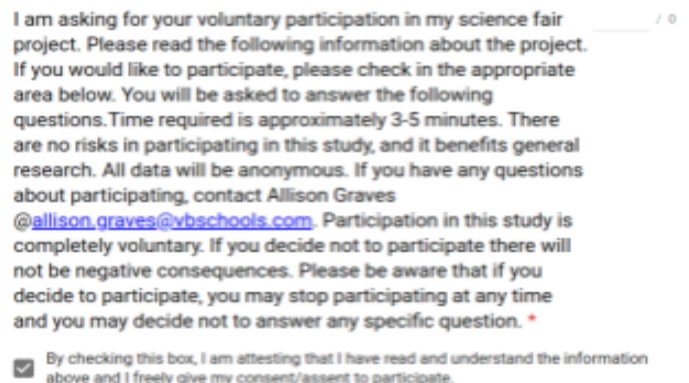
Methodology

Survey Creation

The first and obvious realization was that there had to be a tool used to determine which model fit samples the best. Taking inspiration from the Racial Resentment survey created by the American National Election Studies (ANES), a survey was created lined with questions that demonstrated aspects and assumptions of each model. Through the use of google form, questions were assembled into a ten section survey. It began with a section of human informed consent and was followed by a personal evaluation of political ideology/party and demographics. The following three sections were composed of hypothetical scenarios involving made-up candidates. In each scenario, the candidate demonstrated ideology that aligned either with his or her party, neither specific party (moderate), or the opposite party. It ended with asking them whether or not they understood policies having to do with certain issues that may align with their ideology and then a basic policy question about that issue. There was an additional question asking whether people believed the far-right and far-left would lead to political demise and asked if they agreed or disagreed on a scale of 3.

Sample Selection

An adequate sample size would be that which was larger than 30 samples in order to



I am asking for your voluntary participation in my science fair project. Please read the following information about the project. If you would like to participate, please check in the appropriate area below. You will be asked to answer the following questions. Time required is approximately 3-5 minutes. There are no risks in participating in this study, and it benefits general research. All data will be anonymous. If you have any questions about participating, contact Allison Graves @allison.graves@ybschools.com. Participation in this study is completely voluntary. If you decide not to participate there will not be negative consequences. Please be aware that if you decide to participate, you may stop participating at any time and you may decide not to answer any specific question. *

☒ By checking this box, I am attesting that I have read and understand the information above and I freely give my consent/assent to participate.

Figure 1: Human Informed Consent checked by every participant

help reduce bias and underrepresentation. From a previous experiment, it became known that informed consent forms were one of the biggest inhibitors of collecting samples. People simply did not like to turn them in. To combat this, only people 18 years old or older would be allowed to fill out the survey, and they would only need to check a box to assert their consent shown in figure 1. Given 18 years old is the voting age, it also helped decrease bias by eliminating samples who cannot actually vote.

Originally five recreation centers were chosen to represent the population of Virginia Beach, but ultimately four allowed the conduction of a survey so 12 samples were collected from each recreation center spread across Virginia Beach. Before the survey was conducted, each participant was given a summary of the purpose of the survey without revealing much to sway the responses. Each was also assured that results would be anonymous and to answer as honestly as possible.

Data Analysis

In order to analyze the data, questions had to be sorted in a way such that the answers represented a certain model. The demographics gave a background of a participant and gave a better understanding of why might one vote the way they do. If voted as expected due to demographics, this may serve as evidence of a Michigan model voter. Put to the test was age and race, demographics that often showed trends in voting with certain ideologies and with a certain party ("What Factors Shape Political Attitudes?", 2012). The ideology and party identity questions held the most weight and had to be compared to the answers that followed. As stated before the ideology and party are expected to align with each other and the candidate of their own party in each scenario. Alignment is crucial for the survey in order to allow the assumption

that support of a certain group of ideologies aligns with a certain party (Liberal/Democrat, Moderate/Independent, Conservative/Republican).

The mean was calculated of self-reported knowledge, answers correct on the policy questions, and distrust of far-left and far-right. The chi-square test was used to analyze the association between categorical variables. For each Chi-Square test, the null hypothesis was stated as no association between variables and the alternative hypothesis the opposite.

$$\textbf{Chi-Square Formula: } X^2 = \sum (\text{O}-\text{E})^2/\text{E}$$

$$\textbf{Degrees of freedom: } (r-1)(c-1)$$

The alpha was the standard 0.05 and the degrees of freedom varied and calculated as needed. The Chi-Square table was used to determine a critical value used to possibly reject the null hypothesis. The X^2 was calculated with the Chi-square formula and used the Chi-Square table.

Results

Sociological Influences

The resulting sample size was forty-eight (48) participants of three different races, various ages, party identifications, and ideologies. The sample meant to represent the population of Virginia Beach bared many similarities. The survey distribution of race was reported as 75% white (including Hispanic), 18.5% black, and 6.5% Asian compared to the Virginia Beach distribution of roughly 72% white (including Hispanic), 19.6% black, and 7.7% Asian ("Demographics", 2019). No other races were recorded in the survey. The association between

race and party identity at first seemed evident. A large majority of black and Asian participants were Democrats or Independent while the majority of white participants were Republican or Independent.

Figure 2: Race/party contingency table

		Party Identity						Expected Values (Chi-Square)			
		Democrat	Independent	Republican	Total			Democrat	Independent	Republican	Total
Race	Black	6	2	1	9	Race	Black	3.1875	2.4375	3.375	9
	Asian + other	2	1	0	3		Asian + other	1.0625	0.8125	1.125	3
	White	9	10	17	36		White	12.75	9.75	13.5	36
	Total	17	13	18	48		Total	17	13	18	48
Ho= no association between party ID and race						alpha= 0.05 $X^2=8.5376$					
Ha= association between party ID and race						df= 4 X^2 is < CV so we fail to reject null hypothesis					
						CV= 9.488 * arguably marginal, can't test partial correlation					

However, using the Chi-Square test, the X^2 fell short of the critical value deemed to determine if the null hypothesis could be rejected.

Age was the second demographic chosen to represent a sociological influence in party identity. The age groups ranged from 18-34, 35-49, 50-64, and 65+. As priorly stated, those under the age of 18 were excluded as they cannot legally vote. The ages of those who partook in the survey did resemble the Virginia Beach population as closely as race did. The median age of the population of Virginia Beach is 36.9 and the median age of survey participants fell in the 35 to 49 range ("Demographics", 2019). However, the 35-49 age group was represented almost twice as much in the survey and the 50-64 age group also overshot in representation that category as well. Most survey participants over 50 identified with the Democratic party. The Chi-Square test revealed there was not a significant amount of association between party identity

and age group when tested with the standard alpha value of 0.05.

Figure 3: Age group/party identity contingency table

		Party Identity				Expected Values (Chi-Square)		Party Identity			
		Democrat	Independent	Republican	Total			Democrat	Independent	Republican	Total
Age Group	18-34	6	4	4	14	Age Group	18-34	4.958333333	3.791666667	5.25	14
	35-49	2	8	8	18		35-49	6.375	4.875	6.75	18
	50-64	6	1	4	11		50-64	3.895833333	2.979166667	4.125	11
	65+	3	0	2	5		65+	1.770833333	1.354166667	1.875	5
	Total	17	13	18	48		Total	17	13	18	48
		Ho= no association between party ID and age				alpha=0.05	X ² =10.4345				
		Ho= association between party ID and age				df= 6	X ² is < CV so we fail to reject null hypothesis				
						CV= 12.592					

Alignment and The Vote

Alignment was a crucial aspect that needed to be proven in order to even consider part of the results of the survey. Alignment, or polarization, of political party and ideology, is the simple concept that those of a certain party share certain ideologies. At a glance, many liberals identified as Democrats and many conservatives as Conservatives. In only one instance did a liberal identify as a Republican and upon further investigation, there is strong evidence that this was simply an error as they voted for liberal ideologies and Democratic candidates in the following questions.

A contingency table consisting of party identities (Democrat, Republican, and Independent) and ideologies (liberal, conservative, and moderate) was tested using the Chi-Square method and the resulting X^2 of 30.36 was well above the critical value of 9.488 allowing the rejection of the null hypothesis as expected, allowing for the next part of the survey to have a chance at validity.

The vote was the biggest and most blatant indicator of party-line voting or rational voting. Though three hypothetical scenarios were present in the survey, one was the most important. A Democratic candidate and a Republican candidate were both presented with policies that represented the opposite ideologies standard for the party.

The association between how a participant “voted” for a candidate and party was statistically analyzed twice, once with each party identity and then a second time with attachment to a party (being either a Democrat or Republican versus being an Independent). Both Chi-Square test resulted in insignificant X^2 values and the null hypothesis was rejected, meaning no significant association was recorded. Including moderate Independents and entertaining the assumption they all voted ideologically, 36 out of 48, or 75%, of participants voted with their ideology or against their party. Not including moderate Independents, but including moderates who identified with a party or Independents who had certain ideologies, 27 out of 39 participants voted with their ideology or against their party.

Self-Reported and Demonstrated Knowledge

The end of the survey consisted of six questions asking participants whether or not they agreed with statements on a scale of 3. The first question asked if one would agree that the far-right and far-left will lead to political demise. The average answer to this question was 2.521 on a scale of 3. The following self-reported knowledge questions asked how aware the participant was of key policies and issues including minimum wage, gun control, welfare, immigration, and taxes. The mean response out of a scale of 3 was surprisingly the exact same as the average answer to the previously mentioned question on political demise, 2.521 (this was verified multiple times, as the probability of getting this result out of 48 samples is relatively

low). Finally, the last five questions of the survey were basic questions about the policies participants were asked to answer. A case of overconfidence was evident as, on average, a participant got 2.396 questions correct out of 5, not even half.

Discussion and Conclusions

The results of the study supported the alignment of party identity and ideology and not much else. Alignment helped determine if an independent with a certain ideology leaned towards a certain party and thereby whether or not their vote matched their “party” (only Republican and Democrat candidates were used in the scenarios in order to force a decision). For example, an independent with liberal ideology may tend to lean towards the Democratic party and one could still determine that although they don’t identify with a certain party if they voted for a Democrat with conservative ideology, it would not be rational. When a survey participant chose the candidate with a shared party as opposed to the candidate with the shared ideology, they fell victim to their own psychological attachment to party, supporting the Michigan Model. If a participant identified with an ideology or party on either side of the spectrum (meaning they were a liberal or conservative and/or a Democrat or Republican), one could determine whether they voted ideologically or not. If one identified as a moderate AND an Independent, things became a bit more complicated. One could assume that because of the moderatism and the failure to identify with the two main political parties, these moderate Independents had to vote ideologically. Nonetheless, data shows that most “Independents” actually tend to lean toward one party or the others (LaLoggia, 2019). This is why two contingency tables were made comparing party identity with the Democratic, Independent, or Republican party versus how a participant voted, as well as identifying with a party at all and how a participant voted.

The insignificant association between race and party identity as well as between age group and party identity did not support sociological influence over one's identity with a party. Marginal error is arguable for the association between race and party identity as the X^2 value was less than 1 away from the critical value. Additionally, different races time and time again have shown trends in party lining. Marginal error aside, the acceptance of a false null hypothesis demonstrated type II statistical error. As for the age group, the older age groups had unexpected results, as many voted Democratic shattering the perceived trend of older=conservative. Nonetheless, the generational gap in politics, parties, and ideologies is just recently widening and apart from the millennials, recent data shows that the older generations tend to be only *slightly* more conservative as opposed to the larger gaps seen in gender and race, and the results partially backed this claim ("A Deep Dive Into Party Affiliation", 2019).

Neither party identity nor identifying with the two main political parties was statistically supported as having an effect on how people vote. The large gap between self-reported knowledge and actual knowledge demonstrated overconfidence in participants and an overall lack of adequate knowledge. Many are in agreeance, however, that the far-right and far-left will cause political demise. This would support the Downs Model that inept extremism would fall to moderatism, however, in reality, while most of the population is moderate, extremism continues to control the political sphere (Jenke & Huettel, 2016). Ultimately, the original hypothesis stating that the Michigan Model would prove to be more valid than the Down-Spatial model was rejected in this survey. While overall knowledge seemed to be shown poorly, the vote showed a lack of party-line voting and the contingency tables failed to prove a significant association.

Limitations and Future Improvements

While the hypothesis was rejected and the Downs's Spatial Model proved to be more accurate in this sample, it can't represent the population, in this case, Virginia Beach, due to underrepresentation and response bias. In order to represent the 349,083 Virginia Beach citizens over the age of 18, a sample larger than 48 would need to be used for more accurate results. Due to financial and time limitations, a stratified sample was the most realistic option for data collection.

Alongside underrepresentation was the very evident response bias. Many people would like to believe they vote rationally and would be offended if accused otherwise. In this survey, despite the measures taken to reduce response bias, many participants caught on fast, one even asking if the survey was observing voting by party lines or issues. Additionally, because the survey was only a general model, it could not develop a scenario of significant detail that would cause one to vote for or against one's party. For example, one scenario may be enough to have someone vote along party lines, but a little bit more background in the same scenario would turn someone away from ideological voting.

Another argument would be the simple policy questions at the end. One could argue that because someone doesn't know a single question about a policy, it doesn't mean they are uneducated on the topic. The logic in choosing the questions was that they touched on the essential parts of the policy they asked about. For example, if someone is educated on abortion issues, they should know the federal law regarding abortion from the most popular abortion case in U.S. history (*Roe v. Wade*). Similarly, if one claims to be educated about taxes and minimum wage, they should know how a tax bracket works and what the minimum wage is. Nonetheless, if the survey were repeated, a change to the scale of self-reported knowledge from out-of-3 to

out-of-5 in order to match how many questions there are in the section that follows and get a better comparison would be necessary.

Literature Cited

Peer-reviewed and Scholarly references

- Farber, H. (2010). Rational Choice and Voter Turnout: Evidence from Union Representation Elections. doi: 10.3386/w16160
- Jenke, L., & Huettel, S. A. (2016). Issues or Identity? Cognitive Foundations of Voter Choice. *Trends in Cognitive Sciences*, 20(11), 794–804. doi: 10.1016/j.tics.2016.08.013
- LaLoggia, J. (2019, May 15). 6 facts about U.S. political independents.
<<https://www.pewresearch.org/fact-tank/2019/05/15/facts-about-us-political-independents/>>
- Peele, G. (2004). Governing the Uk: British Politics in the 21st Century (Vol. 1). Wiley.
- Stokes, D. E. (1963). Spatial Models of Party Competition. *American Political Science Review*, 57(2), 368–377. doi: 10.2307/1952828
- Vik, O. (2013). Electoral Implications of the Rational Choice Theory. doi: 10.13140/RG.2.1.3889.7363
- Voting Behavior. (n.d.).
<<https://www.icpsr.umich.edu/icpsrweb/instructors/setups/voting.jsp>>

Non-peer-reviewed references

- A Deep Dive Into Party Affiliation. (2019, December 31).
<<https://www.people-press.org/2015/04/07/a-deep-dive-into-party-affiliation/#party-id-by-race-education>>
- Demographics. (2019).
<<https://www.yesvirginiabeach.com/Business-Environment/Pages/Demographics.aspx>>
- Political Science: Definition of Political Science by Lexico. (n.d.).
<https://www.lexico.com/en/definition/political_science>
- What Factors Shape Political Attitudes? (2012).
<<https://www.ushistory.org/gov/4b.asp>>